Euclid MER VIS galaxy geometry outcomes

Current results 20 October 2026
Adaptive Photutils pipeline with image-based centre checks

All 2777 cutouts have been reprocessed with the adaptive v2 fitter. It returns disc geometry for 2773 galaxies and fails for 4. It recovers 567 of the previous 570 failures. The final recovery pass includes visual exclusions for a foreground-star switch and an unresolved core. A returned result is provisional, not proof that the selected light contours trace a physical disc or bar.

Current measurements

Result	Galaxies	Share of all 2777
Disc geometry returned	2773	99.9%
No counted geometry flags	300	10.8%
Returned geometry with flags	2473	89.1%
Fit failed	4	0.1%

Disc and bar outcomes are separate

Bar outcome	Galaxies
no_bar_detected	453
detected_A	2089
detected_B	231
not_measured	4

A missing accepted bar no longer discards a usable disc result. Bar PA is left blank and no_bar_detected is raised for bar-analysis review. Class A/B labels describe the ellipse-profile signature, not independent visual confirmation. A weaker accepted bar is labelled B and receives a quality flag. The old unstable-bar-PA category is not carried forward as an accepted bar: unstable plateaus are rejected by the new bar test.

Updated report collection

Output	Count
PNG reports	2777
QI reports	2491
NoQI reports	286
PDF batches	47
PDF pages	464

Each PNG includes native pixels, ellipses, ellipticity and sky-PA profiles, geometry comparisons, recovery method and centre offset. PDFs separate QI and NoQI, six galaxies per page and at most 60 per batch. QI includes a counted geometry issue, a failure or partial requested coverage. Nearly face-on alone does not trigger QI.

What changed and how centres are checked

Stage	Adaptive v2 implementation
Resolution	Native pixels for smaller galaxies; adaptive binning for larger targets.
Centre	Search for a nearby light peak, fit free ellipse centres, and check the median outer-disc centre and its scatter.
Fitting	Multiple starting geometries and masks; a further outer fitting attempt where useful.
Acceptance	Normal-convergence samples first. Iteration-limited samples are used only in explicitly relaxed recovery.
Intervals	Weighted axial PA over adaptive disc and stable bar ranges; labelled outer-disc fallback if needed.
Partial success	Disc geometry can be retained without an accepted bar.

Centre measurements and review thresholds

The catalogue position is a starting reference, not an imposed final centre. Red + marks the fitted disc centre and white x the catalogue centre. The fitted centre is the median position of the selected disc ellipses. The manifest stores both zero-based pixel positions, their angular offset and the scatter of selected disc centres. A bright knot or neighbour can still mislead automated centring; the checks identify cases needing inspection.

Median catalogue-to-fitted centre offset: 0.083 arcsec. Large-offset flag: offset > max(0.3 arcsec, 0.15 r_seg). Unstable-centre flag: radial centre scatter > max(0.3 arcsec, 0.10 r_seg). These are operational screening thresholds, not calibrated astrometric error bars.

The synthetic validation used a galaxy of known centre, axis ratio and PA with an offset catalogue position. It recovered the centre to within 0.01 pixel and PA/inclination to within 0.01°. This validates the coordinate conversion and recovery on that controlled case, not every real galaxy.

Inclination and position angle

The inclination remains i = acos(b/a), with q0 = 0. Both disc and bar PA are observed sky angles from North towards East modulo 180°: North 0°, East 90°. On a North-up, East-left image this is counterclockwise. Each ellipse direction is converted through the local FITS WCS. Weighted axial means avoid averaging 179° and 1° to 90°. The bar is not yet deprojected or rotated in these reports.

Quality flags and informational statistics

Counted issue	Galaxies	Share of returned fits
Small selected bar interval	1447	52.2%
Changing disc ellipticity	1355	48.9%
Unstable disc PA	1273	45.9%
No accepted bar; disc retained	453	16.3%
Small disc	298	10.7%
Disc interval fallback	269	9.7%
Weaker accepted bar signature	231	8.3%
Bar and disc intervals close	211	7.6%
Large centre offset	44	1.6%
Star mask removed in retry	18	0.6%
Relaxed recovery	13	0.5%
Iteration-limited samples	10	0.4%
Centre changes across disc	6	0.2%
Neighbour mask removed in recovery	2	<0.1%
Short accepted profile	2	<0.1%
High inclination	1	<0.1%

Counts overlap. New recovery and selection flags make these counts unsuitable for a direct numerical ranking against the old flags. Relaxed recovery, iteration-limited samples and disc-range fallback explicitly identify results obtained using less restrictive conditions. No accepted bar is a limitation on bar geometry, not necessarily on the disc measurement.

Nearly face on is informational

59 returned fits have b/a > 0.90 (inclination below about 25.8°). This is recorded in codex_information, not codex_quality_flags. Disc PA instability is not raised for these nearly circular discs, because their orientation is poorly constrained while their deprojection is small. Their bars may still be useful for analysis. The original run had 200 such markers; the current number changes because disc axis ratios have been remeasured with different radial selections.

Remaining failures

-586374330510251740: ValueError: Only 0 accepted isophotes after recovery; need at least 3

-524824554283124036: Visual review: broader peak search switched to a saturated foreground star with diffraction spikes; no valid target-disc geometry adopted.

2645436353651870062: ValueError: Only 0 accepted isophotes after recovery; need at least 3

2697242050653408730: Visual review: accepted contours cover only the unresolved central region below 0.5 arcsec; no resolved disc geometry adopted.

Comparisons and using the results

Comparison with Euclid whole-object shape	Median absolute difference	Pairs
Adjusted detection PA	8.05°	2773
Detection inclination proxy	6.52°	2773
Sérsic inclination proxy	5.50°	2773

Euclid detection and single-Sérsic shapes describe whole-object light; they are not independent outer-disc or bar measurements. No independent Euclid bar PA is provided. Inclination proxies retain the q0=0 assumption for like-for-like comparison.

Euclid PA uses the adopted project correction: (WCS-converted catalogue angle +90°) modulo 180°. The WCS step already converts image x to North towards East. The extra +90° is consistent with a y-referenced catalogue angle being interpreted as the documented x-reference, but this explanation remains unconfirmed. The raw catalogue values are preserved.

Comparison with supplied fitted manifest	Median absolute difference	Pairs
disc	2.76°	2773
bar	0.45°	2292
inclination	2.37°	2773

The supplied manifest uses q0=0.2 and different interval/acceptance choices. Our q0 remains zero. These differences therefore include methodological effects and are not formal independent accuracy tests.

Where to find the current outputs

Geometry: Euclid_MER_VIS_geometry/MER_VIS_local_geometry_manifest.csv in the research data folder. Individual reports: MER_VIS_galaxy_reports/all_2777/png_QI and png_NoQI. PDF reports: pdf_QI and pdf_NoQI. MER_VIS_report_review_index.csv gives the PNG, PDF, page and position for each galaxy.

The raw cutout FITS files are unchanged. These reports do not create final deprojected images. Inspect flagged quantities, centres and selected radial ranges before adopting geometry. A reproducibility record retains the original summary manifest as an audit baseline; superseded presentation reports and image/PDF collections have been replaced.

Sources: current MER_VIS_local_geometry_manifest.csv, MER_VIS_v2_summary.json, MER_VIS_report_verification.json and pipeline version MER_VIS_adaptive_v2_2026-10-20. Code: Euclid/adaptive_MER_VIS. The supplied methodology and fitted manifest are retained as comparison references.

Understanding the bar signature classes

What class A and class B mean

These are operational categories defined by our ellipse-fitting code. They describe how clearly the radial ellipse profiles resemble a bar. They are not Galaxy Zoo or Zoobot classes, probabilities of a bar, or measurements of physical bar strength.

The common acceptance test

A candidate must have a smoothed ellipticity peak of at least 0.20 and a plateau of at least three accepted ellipses. The plateau sky PA scatter must be no more than 10°. Beyond the peak, ellipticity must drop by at least 0.08; alternatively, a drop of at least 0.04 is accepted if the outer PA twist or difference from the preliminary outer-disc PA is at least 10°.

Class A means a clearer ellipse signature

An accepted candidate is class A only if all three stronger conditions hold: ellipticity drops by at least 0.10 beyond the peak; plateau PA scatter is no more than 5°; and peak ellipticity is at least 0.30. Ellipticity is $1 - b/a$, so 0.30 corresponds to an ellipse axis ratio b/a of 0.70.

Class B means accepted with weaker evidence

A candidate is class B if it passes the common acceptance test but misses at least one class A threshold. For example, peak ellipticity 0.35, drop 0.08 and PA scatter 4° passes the common test but is B because the drop is below 0.10. Peak ellipticity 0.40, drop 0.15 and scatter 7° is also B because scatter exceeds 5°.

How to interpret the quality flag

The flag weak_ellipse_bar_signature marks every accepted class B candidate for review. Class B does not mean that the galaxy definitely has a physically weak bar. A bulge, ring or spiral arm can mimic a bar profile, including a class A profile. Inspect the image and selected ellipses before adopting the bar PA. If no candidate passes the common test, no bar PA is adopted; the disc measurement may still be retained.